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Greater Noida
Autonomous Institute

Project Presentation on **LEARNIFY**



By

Project Group No.: 5

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- Motivation
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Learnify is:

- **Learnify** is an online learning and collaboration platform.
- It allows users to **learn, connect, and interact through live video sessions.**
- The platform combines **education with real-time communication tools.**

Objectives

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- To develop a real-time online learning platform with low latency
- To enable multi-user video and audio communication using WebRTC
- To improve student engagement through chat and reaction features
- To provide secure user authentication and data protection
- To design a scalable and lightweight system architecture
- To track and analyze learner engagement for academic research

- During online classes and remote learning, many students faced difficulties in **interaction and engagement**.
- Traditional video platforms often feel **complex or distracting** for learners.
- There was a need for a **simple, student-friendly platform** focused on learning.
- Learnify was created to **bridge the gap between teachers and students** through real-time communication.
- The goal was to make **online learning more interactive, accessible, and effective**.



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Literature Survey

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S. No.	Title	Year	Author	Publication	Remarks
1	Home Security System	2021	P. Amith Teja A. Anne Frank Joe V. Kalist	2021 International Conference on	An intruder activity is sensed based on the information received from the installed sensors in
2	IoT Based	2020	Sonali Das Dr. Neelananarayan	2020 IJESC Volume 10	The camera continuously checks the status of that particular place by camera and sensor. Person is

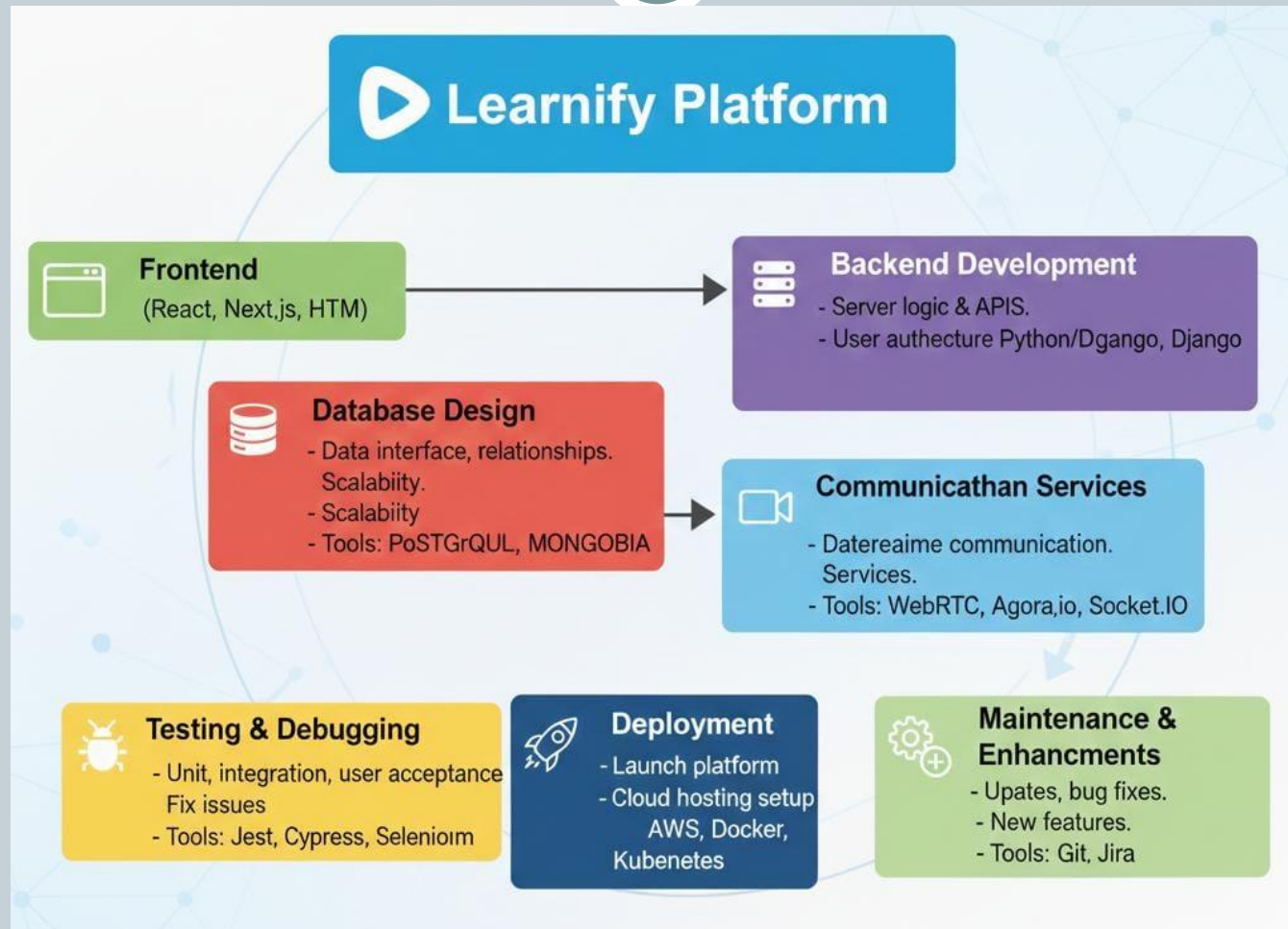


Literature Survey

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S. No.	Title	Year	Author	Publication	Remarks
3	System	2021	Sanjana Kute Ruthvik Pimpalkar Vaibhav More M.R.Gobal	IRE Journals Volume 4 Issue 10 APR 2021	In this framework, we utilize a camera alongside raspberry pi alongside a circuit with LCD show IR
4	IOT Based Anti-Theft	2020	D. Pooja Sri N. Gayathri K.Heshma	7th International Electronic Conference on	In this system, secure flooring tile connected with IOT,

- Design and develop **Learnify**, a real-time learning and video meeting platform.
- Implement **video, audio, chat, and screen sharing** using modern web technologies.
- Provide **secure user authentication** for controlled access.
- Focus on a **responsive, user-friendly interface** for students and educators.
- Enhance **interaction and collaboration** in online learning environments.





Frontend

- Developed using **React.js** for a responsive and interactive user interface.
- Uses **Material UI and CSS** for modern and user-friendly design.
- Handles user authentication, video UI, and real-time interactions.

Backend

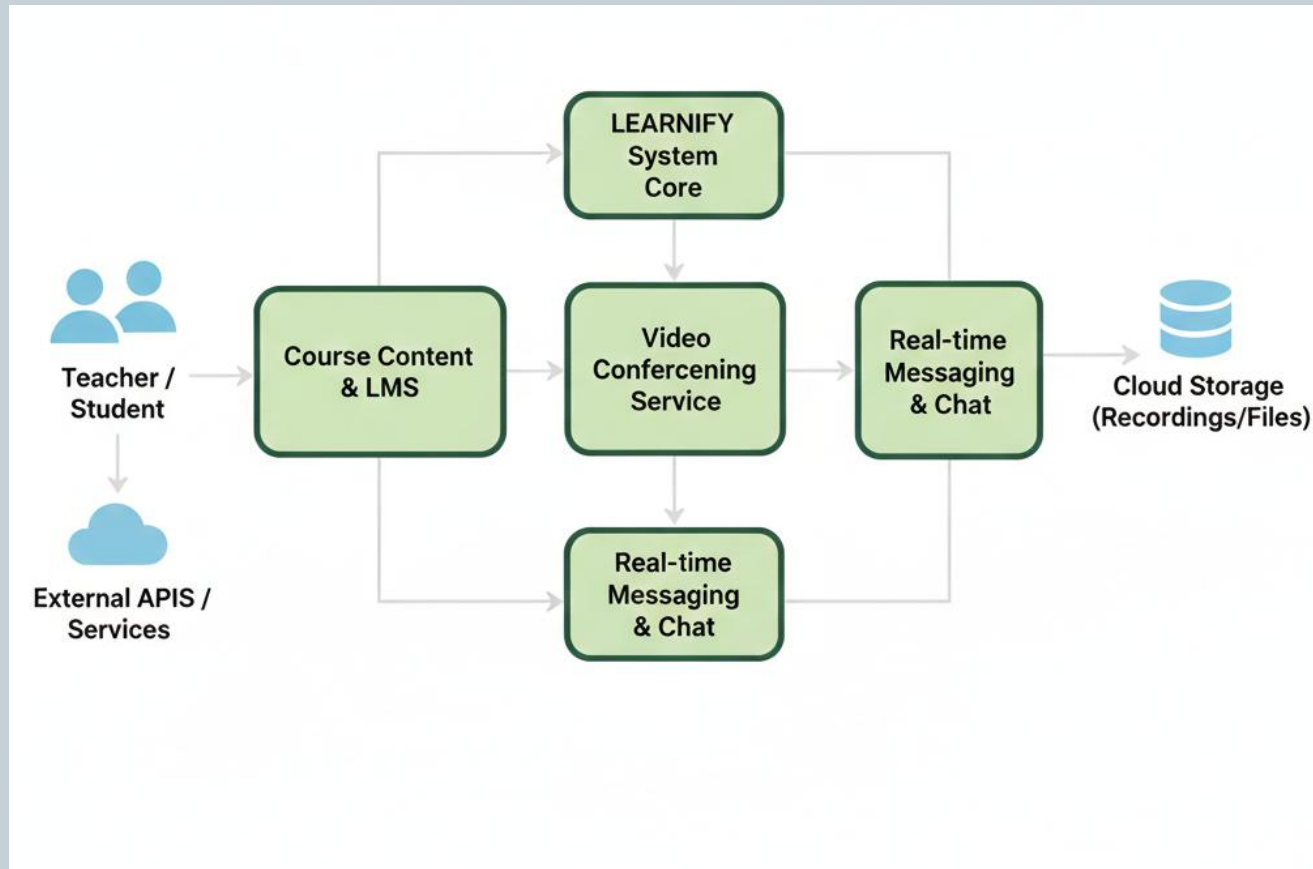
- Built with **Node.js and Express.js** to manage APIs and application logic.
- Uses **Socket.io** for real-time communication and signaling.
- Implements authentication and meeting management services.

Database

- Uses **MongoDB** to store user information and meeting data.
- Ensures secure and efficient data storage and retrieval.

Block Diagram

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- Learnify provides a **simple and effective solution** for real-time online learning.
- The platform enables **secure, interactive communication** through video, audio, and chat.
- It enhances **remote education and collaboration** using modern web technologies.

- Implement **face recognition-based attendance system** for automated tracking.
- Implement **AI-based features** like automatic attendance tracking and meeting summaries.
- Add **recording and playback functionality** for recorded lectures.
- Introduce **screen annotation and whiteboard tools** for better interaction.
- Integrate **learning management features** such as assignments, quizzes, and progress tracking.

- [1] 2021 International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE) | 978-1-7281- 7741-0/20/\$31.00 ©2021 IEEE | DOI: 10.1109/ICACITE51222.2021.9404551
- [2] Priya B. Patel, Viraj M. Choksi, Swapna Jadhav, M.B.Potdar, “Smart Motion Detection System using Raspberry Pi” International Journal of Applied Information Systems (IJ AIS) – ISSN: 2249-0868 Foundation of Computer Science FCS, New York, USA Volume 10 – No.5, February 2016.
- [3] IOT based Raspberry Pi Home Security System with Email Alert, Circuit Digest.com, and Saddam.
- [4] Real Time Video Monitoring System Using Raspberry Pi “Sameer Gode, Devendra Hande, Atharva Hinge, Madhuri Ghuge” International Journal of Scientific Research and Engineering Development-- Volume2 Issue 2, Mar –Apr 2019
- [5] IOT Based anti-theft detection and alerting system using Raspberry PI “U. Sirisha, D. Poojasri, N. Gayathri, K. Heshma, G. Raja Sekhar” International Research Journal of Engineering and Technology (IRJET) 07 March,2020.
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- [7] IOT Based Advance Security System by Using Raspberry PI “Nagula Shyam Kumar, Nivedita.M” Internation Journal & Magazine Of Engineering, Technology, Management and Research



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*Thank
you*

